

Atilla Saadat

Los Angeles, CA, USA

U.S. Permanent Resident (EB-1A, Extraordinary Ability)

 atillasaadat.com  contact@atillasaadat.com  [LinkedIn](#)

 [ResearchGate](#)  [Google Scholar](#)  [Github](#)  [Devpost](#)

Industry Experience

Spacecraft Guidance, Navigation, & Control (GNC) Engineer II

Nov 2025 – Present

Varda Space Industries

Los Angeles, CA

- Served as GNC Responsible Engineer for W-Series 8 and W-Series 9, architecting next-generation reentry vehicle GNC capabilities for multi-vehicle operations, mission-agnostic FAA flight safety analysis, increased mission cadence, and greater flight autonomy
- Led Monte Carlo analysis, trajectory design optimization, and on-orbit GNC operations for [W-Series 4](#), and post-flight reconstruction for the [successful launch and reentry of W-Series 5](#)

Spacecraft Guidance, Navigation, & Control (GNC) Engineer

Oct 2022 – Nov 2025

Turion Space Corp.

Irvine, CA

- Led GNC engineering for [Droid.001](#) and [Droid.002](#) from concept through launch, commissioning, and on-orbit operations, following deployment on SpaceX [Transporter-8](#) and [Transporter-13](#). Developed verification and validation (V&V) methods spanning pointing budgets, power generation analysis, on-orbit diagnostics, and flight-data validation, achieving $<0.05^\circ$ (2σ) pointing accuracy on Droid.001 for Space Situational Awareness (SSA) and $<6''$ (2σ) on Droid.002 for Non-Earth Imaging (NEI).
- Designed and implemented a Python-based autonomous NEI target tracking and mission-planning algorithm for inter-satellite imaging, optimizing attitude profiles while enforcing Sun/Earth exclusion constraints and enabling [1,022 successful tasking collects and 26,467 high-precision images](#) across Droid.001 and Droid.002.
- Built high-fidelity spacecraft simulations with Hardware/Software-in-the-Loop (HITL/SITL) capabilities to validate autonomous inter-satellite imaging operations, evaluate mission design trades, and prototype new concepts of operations. Conducted Monte Carlo success analyses and calibrated body-frame imagers in orbit using star field imagery to ensure sub-arcminute NEI pointing alignment.
- Implemented batch least-squares orbit determination from on-orbit multiplicative extended Kalman filter (MEKF) state estimates, uploading refined ephemeris to Space-Track for conjunction analysis and maneuver planning. Verified LEO ADCS torque/momentum authority and executed power-compliant pointing campaigns, advancing GNC autonomy across Turion's SSA and NEI mission portfolio.

Hardware Functional Safety Engineering Intern

May 2022 – Sept 2022

NVIDIA Corp.

Santa Clara, CA

- Automated Fault Tree Analysis (FTA) across over 100,000 failure modes in Python, deployed in production for the [Mercedes-Benz × NVIDIA Jetson AGX Orin system-on-chip \(SoC\)](#) under [ISO 26262](#) (Road Vehicles – Functional Safety)
- Streamlined failure analysis workflows for autonomous vehicle development, improving the traceability and repeatability of functional safety evidence across hardware fault campaigns

Robotic Controls Engineering Intern II

Sept 2019 – Jan 2020

Mujin Inc.

Tokyo, Japan

- Developed a torque model validation algorithm with tailored metrics and fault detection, improving torque coefficient estimate confidence by over 70%
- Extended and productionized the dynamics identification technique for the Mujin Controller, optimizing torque model coefficient calculations for production robots across Mujin's industrial fleet
- Collaborated with Mujin's legal department to file [two proprietary software patent applications](#), active in the USA and Japan and pending in China

Robotic Controls Engineering Intern I

Mujin Inc.

May 2017 – May 2018

Tokyo, Japan

- Designed a dynamics identification technique for the Mujin Controller, computing friction, center of mass, and inertia tensor coefficients to enhance robot torque modeling accuracy
- Developed and deployed production-grade code validated against experimental results, improving robot trajectory tracking accuracy under high-acceleration scenarios by over 85%
- Integrated robot armature trajectory generation with data analytics and custom UX/UI visualizations, streamlining operator workflows and improving system observability

Space Systems Engineering Intern (Lunar Exploration)

Canadensys Aerospace Corp.

May 2016 – Sept 2016

Toronto, Ontario

- Led development of a remotely-operated lunar rover prototype integrating stereo, fisheye, and 3D camera modalities on a \$10k budget, supporting lunar surface exploration objectives
- Designed and integrated a ground station GUI combining real-time camera feed visualization with vehicle command and control, improving operational efficiency for remote rover operations

Education

M.S. – Computer Science

Georgia Institute of Technology

Jan 2025 – May 2027

Atlanta, Georgia

- Specialization: Artificial Intelligence; coursework and projects emphasize deep learning, reinforcement learning, robotics AI, and space computing for autonomous aerospace systems
- Research: Adaptive F Prime telemetry for resource-constrained spacecraft (AIAA SciTech 2027) and Vision-Language-Action models for collaborative In-Space Servicing, Assembly & Manufacturing (ISAM) at the [Aerospace Robotics Laboratory](#)
- Research Advisors: [Dr. Ada Gavrilovska](#) & [Dr. Yashwanth Kumar Nakka](#)

M.A.Sc. – Aerospace Science & Engineering

University of Toronto

Sept 2020 – Aug 2022

Toronto, Ontario

- Spacecraft Systems Engineering and Attitude Determination & Control Systems (ADCS) research and development at the [UTIAS Space Flight Laboratory \(SFL\)](#)
- Thesis: [Design and Test Optimizations for Spacecraft Attitude Determination and Control Subsystems](#)
- Research Advisor: Dr. Robert E. Zee
- Awarded Research Fellowship & Internal Student Fellowship. Graduated with “High Distinction”

B.A.Sc. – Honours Mechanical Engineering w/ Aerospace Option

University of Windsor

Sept 2015 – Aug 2020

Windsor, Ontario

- Aerospace ML Researcher at the [Intelligent Control, Analysis, and Modeling \(iCAM\) Lab](#)
- Research Advisor: [Dr. Afshin Rahimi](#)
- Dean's List (3x)

Technical Skills

Research Areas: Spacecraft GNC, ADCS, state estimation, numerical optimization, trajectory optimization, orbit determination, autonomous mission planning, embedded spacecraft systems, CubeSats, distributed space systems, robotics, machine learning

Languages: Python, MATLAB/Simulink, C/C++, Java, JavaScript, PHP, HTML/CSS, NASA JPL F Prime

Engineering Tools: AGI STK, FreeFlyer, GMAT, Basilisk, OpenC3 COSMOS, SolidEdge, Siemens NX w/ Space Systems Thermal, SolidWorks, Catia V5, KiCad, Altium

ML/Scientific Computing: NumPy, SciPy, Matplotlib, Plotly, Scikit-learn, OpenCV, TensorFlow, Keras, TSfresh, Hyperopt, pandas, PyTorch

Developer Tools: Linux, Git, Bash, Docker, PyTest, Google Cloud Platform, AWS, MongoDB, MySQL, PostgreSQL, VS Code, Eclipse, Android Studio, $\LaTeX$

Fabrication and Lab: 3D printing, MIG/TIG/Stick welding, wire harnessing, multimeters, oscilloscopes, ESD compliance, SMT soldering, Raspberry Pi/Arduino/BeagleBone

Q Research Experience

Graduate Researcher

Sept 2020 – Sept 2022

UTIAS Space Flight Laboratory

University of Toronto

- Automated ADCS hardware testing (rate sensors, magnetometers, magnetorquers, fine sun sensors, reaction wheels) across 16 satellites in the [HawkEye 360 constellation](#), reducing test time by 63% through Python software and hardware workflow redesign, producing repeatable V&V methods, improved data quality, and stronger flight-readiness evidence
- Developed a novel Earth horizon sensor using a repurposed spacecraft inspection camera, achieving 1° RMSE nadir-vector estimation accuracy via Python/OpenCV image processing and validating the method against STK EOIR simulation and on-orbit NorSat-2 imagery. Presented at the [2022 Small Satellite Conference](#), winning **1st place** in the student competition
- Developed an in-situ inertia tensor estimation method for on-orbit satellites, validated through MATLAB/Simulink ADCS simulation, and implemented new reaction-wheel attitude commands in C++ flight code for the HawkEye 360 mission. Related conference paper selected by the Canadian Space Agency (CSA) as one of two Canadian delegate submissions to the [IAC 2022 E2 Student Conference](#)
- Built a model-based solar power generation analysis tool in Python and STK for [NASA MSFC's StarBurst mission](#), incorporating photovoltaic self-shadowing effects from the SolidEdge CAD model. Automated analysis across 135+ attitude and orbital cases with shadowed string thresholding, producing high-fidelity power budget reports

Undergraduate Researcher

Mar 2019 – Oct 2020

Intelligent Control, Analysis, and Modeling (iCAM) Laboratory

University of Windsor

- Developed a novel ensemble-based ML algorithm for fault detection in satellite reaction wheels, using scikit-learn, TensorFlow, and Keras for fault case classification and feature extraction
- Conducted time-series data analysis research, optimizing dataset generation and establishing best practices for feature extraction in ML-based fault diagnosis applications
- Co-authored published papers and conference presentations advancing the field of intelligent control and fault diagnosis for spacecraft systems

🏆 Awards

[SME 30 under 30 Award](#) | *Society of Manufacturing Engineers*

Oct 2026

- Recognized as an emerging leader in manufacturing and engineering for pioneering contributions to spacecraft guidance, navigation, and control

[4th Place in MIT STORM-AI Challenge](#) | *Massachusetts Institute of Technology (MIT) ARCLab*

Jul 2025

- Built a metric-aligned BiGRU-Attention model for real-time 72-hour thermospheric density forecasts, exceeding transformer baselines at millisecond latency and demonstrating ML-enabled autonomy for space situational awareness and onboard planning

[Payload Pioneer 30 under 30 Award](#) | *Payload Space*

Oct 2024

- Recognized by Payload's "30 under 30" for achievements across the space industry, including innovation, advocacy, and leadership

[SSPI 20 under 35 Award](#) | *Space & Satellite Professionals International (SSPI)*

Sept 2024

- Recognized by SSPI's "20 under 35" for outstanding contributions and leadership in the space industry

[1st place in Frank J. Redd Student Competition](#) | *Small Satellite Conference*

Aug 2022

- Awarded \$10k for [best paper](#) and presentation in an international student competition

[Canadian Delegate in IAC Student Conference](#) | *Canadian Space Agency (CSA)*

Aug 2022

- Selected as one of two students across Canada to attend the IAC 2022 50th Student Conference by the CSA

[Research Fellowship & Internal Student Fellowship](#) | *University of Toronto*

Sept 2020

- Awarded fully-funded tuition and stipend for admittance into [UTIAS Space Flight Laboratory](#)

[1st place team in CSDC Critical Design Review](#) | *Canadian Satellite Design Challenge (CSDC)*

Oct 2019

- Presented CDR-level designs for a 3U Cube Satellite to CSA, MDA, and ABB spacecraft engineers

[Ontario Aerospace Council Scholarship](#) | *Ontario Aerospace Council*

May 2019

- Awarded one of four awards to Ontario aerospace engineering students for academic and extra-curricular achievements related to aerospace studies

[Dean of Engineering Conference Grant](#) | *University of Windsor*

Jan 2019

- Awarded directly by the Dean of Engineering to attend the ESA Atlantic from Space Workshop

ESA CryoSat-2 Science Meeting Student Award | *Canadian Space Agency (CSA)* **Mar 2017**

- Awarded to select students across Canada to attend ESA CryoSat-2 meeting as student representatives for the CSA, used for research and resources for [CryoRoute](#)

P.E.O. Windsor-Essex Chapter Bursary | *Professional Engineers of Ontario (P.E.O)* **Mar 2016**

- Awarded to 1st year engineering students who have demonstrated excellent academic achievement

University of Windsor Scholarships & Bursaries

\$16k total – Awarded for undergraduate academic achievement and extra-curricular activities, unless otherwise stated:

- Student Access Bursary | Jun 2020 & Jun 2019
- Ontario First Generation Bursary | Jun 2019 & Sept 2018
- UWindsor In-Course Bursary | Jun 2019
- Student Life Enhancement Fund | Jan 2019 | Awarded to attend ESA Atlantic from Space Workshop
- Helen Norma Laframboise Scholarship | Sept 2018
- Faculty Retirees' Scholarship | Sept 2018
- Ontario Student Opportunity Trust Fund (OSOTF) Scholarship | Sept 2018
- Marian McLean Campus Spirit Scholarship | Sept 2018 | Awarded for WinSAT events held on campus
- International COOP Bursary | Jan 2018 | Awarded for Mujin Inc. internship in Japan
- Duronio Family Scholarship | Jan 2017
- University of Windsor Entrance Scholarship | Sept 2015
- Golden Key International Honour Society | Sept 2016 | Nomination

Master's Thesis

Saadat, A. (2022). *Design and Test Optimizations for Spacecraft Attitude Determination and Control Subsystems*. Master's thesis, University of Toronto Institute for Aerospace Studies, Space Flight Laboratory. [doi:10.1807/125667](https://doi.org/10.1807/125667)

Peer-Reviewed Journal Articles

Rahimi, A., & **Saadat, A.** (March, 2020). [Fault isolation of reaction wheels onboard three-axis controlled in-orbit satellite using ensemble machine learning](#). *Aerospace Systems*, 3(2), 119–126. doi.org/10.1007/s42401-020-00046-x

Conference Proceedings & Presentations

Saadat, A., V. Bhosale, K. Bhardwaj, A. Gavrilovska (January, 2027). [An Adaptive Telemetry Probe for Resource-Constrained Spacecraft Using F Prime](#). *AIAA SciTech Forum 2027*, Orlando, Florida. (*Accepted*)

Saadat, A., V. Chen, J. Austin (September, 2025). End-to-End Autonomous Mission Planning and Spacecraft Attitude Optimization for Resident Space Object Imaging. *Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference 2025*, Maui, Hawaii. [\[PDF\]](#)

Saadat, A. Attitude Determination with Self-Inspection Cameras Repurposed as Earth Horizon Sensors. *Frank J. Redd Student Competition - Small Satellite Conference 2022*, Utah State University, Logan, Utah. (1st Place Paper and Presentation) [\[PDF\]](#)

Saadat, A. On-orbit Spacecraft Inertia Tensor Estimation. *International Astronautical Congress 2022*, Paris, France. [\[PDF\]](#)

Folami, F.O., Rahimi, A. & **Saadat, A.** (March, 2020). Fault Isolation of Reaction Wheels on-board 3-axis Controlled in-orbit Satellite Using Machine Learning Techniques. IAC-20.B4.9-GTS.5.10, *International Astronautical Congress 2020*, Dubai (Online), United Arab Emirates. [\[PDF\]](#)

Rahimi, A., & **Saadat, A.** (July, 2019). Fault isolation of reaction wheels on-board 3-axis controlled in-orbit satellite using ensemble machine learning techniques. *The International Conference on Aerospace System Science and Engineering*, Toronto, Ontario. [\[PDF\]](#)

Technical Reports

Saadat, A., Weeratunga, M., Hasan, M., & Mehroke, A. Reconstructing ESA Time-Series Trojans via Fourier-Regularized Deep Learning. Technical Report, CS7643 Deep Learning Final Report, Georgia Institute of Technology, July 2025. DOI: [10.13140/RG.2.2.17228.68480](https://doi.org/10.13140/RG.2.2.17228.68480). [\[PDF\]](#)

Saadat, A. Real-Time Thermospheric Density Forecasting with a Metric-Aligned BiGRU-Attention Network for Space Situational Awareness. Technical Report, MIT ARCLab AI Competition, May 2025. [\[PDF\]](#)

Patents

Saadat, A., & Usui, K. (May, 2020). Method And Computing System For Estimating Parameter For Robot Operation. Patent applications in USA ([Active](#)), Japan ([Active](#)), and China ([Pending](#))

Saadat, A., & Usui, K. (May, 2020). Method And Computing System For Determining A Value Of An Error Parameter Indicative Of Quality Of Robot Calibration. Patent applications in USA ([Active](#)), Japan ([Active](#)), and China ([Pending](#))

Poster Presentations

Saadat, A., V. Chen, J. Austin (September, 2025). [End-to-End Autonomous Mission Planning and Spacecraft Attitude Optimization for Resident Space Object Imaging](#). *Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference 2025*, Maui, HI

Saadat, A. (March, 2019). [CryoRoute - Generating Globally-Optimized Naval Routes in the Arctic Ocean](#). *UWillDiscover Conference*, University of Windsor, Windsor, ON

Saadat, A. (January, 2019). [CryoRoute - Generating Globally-Optimized Naval Routes in the Arctic Ocean](#). *ESA's Atlantic from Space Workshop*, National Oceanography Centre, Southampton, UK

Academic Reviews

Judge for [AIAA Region VI Student Conference 2025](#)

Completed Graduate-level Courses

- | | | |
|---|------------------------------|---|
| • Microsatellite Design I | • Flight Dynamics & Control | • CS-7643 Deep Learning |
| • Microsatellite Design II | • Aerospace Structures | • CS-7638 Robotics: AI Techniques |
| • Spacecraft Dynamics & Control | • Aerodynamics & Performance | • CS-7641 Machine Learning |
| • Computational Optimization | • Aerospace Propulsion | • CS-7642 Reinforcement Learning |
| • Control Theory | • Aerospace Materials & Mfg. | • CS-8903 Space Computing |

Co-Curricular Experience and Projects

[Real-Time Spacecraft Thermospheric Density Forecasting](#)

May 2025

MIT Astrodynamics, Space, Robotics, and Controls Laboratory (ARCLab)

- Developed and open-sourced a PyTorch BiGRU-Attention model that converts 60 days of solar-wind and orbital data into 72-hour, 10-minute-step thermospheric density forecasts in milliseconds, training in $\leq 4\text{h}$ on a single RTX 3090 Ti
- Ranked **4th out of 159 participants** (0.709 normalized skill score, exceeding the transformer baseline) and achieved the highest paper score in the MIT ARCLab STORM-AI Competition, delivering a deployable real-time solution for drag prediction and collision-risk screening

President / Founder / Lead Space Systems Engineer

2018 – 2020

University of Windsor Space & Aeronautics Team (WinSAT)

- Founded WinSAT and led the team to **1st place** in the [Canadian Satellite Design Challenge \(CSDC\)](#) Critical Design Review, directing design and development of a 3U CubeSat for LEO while managing 20+ students and securing over \$40k in funding
- Led end-to-end development of all satellite subsystems including Attitude Determination & Control, RF Communications, and Structural & Thermal, driving cross-disciplinary integration across a multidisciplinary team

- Organized and led educational workshops on satellite engineering and Amateur Radio Operator certification for 30+ students, expanding hands-on technical capacity within the university community

Lead Avionics Engineering - Sounding Rocket 2015 – 2019

Rocketry Payload Division, UWindsor Rocketry Team

- Designed and built a [custom aerodynamics data acquisition module](#) using Raspberry Pi and a multi-sensor array, enabling precise in-flight aerodynamic analysis and performance validation
- Integrated custom PCB design with embedded software and redundancy handling to reliably capture and process high-rate rocket telemetry including acceleration, angular velocity, and atmospheric conditions

CryoRoute - Globally Optimized Naval Routes in the Arctic Ocean Jan 2019

Independent Research Project

- Developed a [novel route optimization tool](#) for Arctic naval navigation, using CryoSat-2 SAR sea ice freeboard data from the [Centre for Polar Observation and Modelling \(CPOM\)](#) to improve safety and transit efficiency
- Implemented a tailored A* pathfinding algorithm minimizing sea ice thickness, distance, and transit duration to produce optimized Arctic shipping routes
- Presented findings at NASA Space Apps Waterloo, UWillDiscover Conference, and ESA's Atlantic from Space Workshop

Inertial Electrostatic Confinement (IEC) Nuclear Fusion Reactor Demonstrator Aug 2015

Independent Research Project

- Designed and built a [vacuum chamber](#) to simulate atmospheric gas ionization and plasma generation, replicating core processes of a deuterium-powered IEC nuclear fusion reaction
- Operated a 15 kV high-voltage power supply under strict safety protocols to maintain stable plasma conditions throughout experimentation
- Characterized the relationship between internal grid geometry and plasma formation rates using visual luminescence techniques, laying groundwork for future neutron radiation simulations and fusion validation

✿ Professional Training & Certification

Introductory and Advanced F Prime Workshop Certification Feb 2026

NASA Jet Propulsion Laboratory Pasadena, CA

Nationally Registered Emergency Medical Technician Sept 2025

National Registry of Emergency Medical Technicians (NREMT) United States

PADI Rescue Diver w/ Emergency First Response (EFR) Oct 2024

Professional Association of Diving Instructors (PADI) Laguna Beach, CA

Basic Life Support (CPR & AED) Sept 2024

American Heart Association United States

PADI Advanced Open Water Diver Apr 2024

Professional Association of Diving Instructors (PADI) Laguna Beach, CA

- Specialties: Deep Dive, Search & Recovery, Peak Performance Buoyancy, U/W Navigation, U/W Naturalist

PADI Enriched Air Nitrox Diver Apr 2024

Professional Association of Diving Instructors (PADI) Laguna Beach, CA

Wilderness Medicine - First Aid / Epinephrine Auto-injector Apr 2024

NOLS Wilderness Medicine Los Angeles, CA

PADI Open Water Diver Feb 2024

Professional Association of Diving Instructors (PADI) Laguna Beach, CA

Canadian Amateur Radio Operator Certificate - Callsign: **VA3UWS** Jan 2019

Innovation, Science and Economic Development Canada Government of Canada

- Basic with Honours - Access to all amateur radio frequency bands

Siemens NX Space Systems Thermal Training Feb 2019

CSDC Structural & Thermal Analysis Workshop Magellan Aerospace, Winnipeg, Manitoba

- Learned spacecraft structural and thermal analysis techniques for small satellites in Siemens NX

Hackathon Projects & Awards

GeoRisk

2017

Winter Hackathon

University of Windsor, Windsor, Ontario

- Developed a Google Maps web-based application to analyze client data for a sponsoring insurance company
- Awards: [UWindsor EpiCentre Startup Series-A Funding \(\\$6k\)](#), later used to seed [WinSAT](#)

Lubdub

2017

SpartaHacks III

Michigan State University, East Lansing, MI

- Developed a web application that records and detects heart defects using audio wave analysis and a stethoscope, achieving real-time diagnostic capabilities
- Awards: SpartaHacks 2nd Place Winner, ThoughtWorks Honorable Mention

Gateway 2.0

2015

Facebook Global Hackathon

Facebook HQ, Menlo Park, CA

- Extended an iOS messaging app with embedded API search query responses and natural language processing

Gateway

2015

MHacks 6

University of Michigan, Ann Arbor, MI

- Developed an iOS messaging app with embedded API search query responses and natural language processing
- Awards: Best Use of Facebook Parse API (Invitation to Facebook Global Hackathon), Best iOS App (sponsored by Apple Inc.), Best Use of Expedia API

Professional Affiliations & Community Engagement

Aerospace Education Officer, 2d Lt

Apr 2024 – Nov 2025

[Civil Air Patrol \(CAP\) - Squadron 232](#)

Mission Viejo, CA

- Deliver STEM education through lectures, hands-on activities, and rocketry/Arduino projects to inspire cadets in aerospace and engineering fundamentals.

Programs Chair

June 2024 – Sept 2026

[American Institute of Aeronautics and Astronautics \(AIAA\) - Orange County Chapter](#)

Orange County, CA

- Coordinate and host distinguished guest speakers to advance aerospace dialogue and professional development within the local AIAA community
- 2025-2026 Outstanding Section Award - 3rd Place, Large Category

Media Coverage

[STARFIRE: FLEET COMMAND](#) | [Turion Space Corp.](#)

Sept 2024

[Droid.001 Alpha Centauri Imaging](#) | [Turion Space Corp.](#)

Jun 2024

[Droid.001 Moon Image Results](#) | [Turion Space Corp.](#)

Apr 2024

[Droid.001 Mission and Simulation Highlights](#) | [Turion Space Corp.](#)

Sept 2023

[Small Satellite Student Competition Award](#) | [University of Toronto Engineering Dept.](#)

Aug 2022

[UWindsor satellite design leading student competition](#) | [University of Windsor Engineering Dept.](#)

Nov 2019

[WinSAT wins critical satellite design review](#) | [Windsor Star – News Outlet](#)

Nov 2019

[Aerospace engineering student aiming high](#) | [University of Windsor Engineering Dept.](#)

Aug 2019

[Mujin Inc. Internship - Student Spotlight](#) | [University of Windsor Co-operative Education Dept.](#)

Oct 2018